Vorlesung 18

Maxwellsche Gleichungen: Skalar und Vektorpotentiale

2. Maxw. Gl.: $\vec{\nabla} \cdot \vec{B} = 0 \quad \Rightarrow \quad \boxed{\vec{B} = \vec{\nabla} \times \vec{A}}$

3. Maxw. Gl.: $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \neq \vec{0} \quad \Rightarrow \quad \vec{E} \neq -\vec{\nabla} \varphi$

aber $\vec{\nabla} \times \vec{E} = -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{A})$

$$\vec{\nabla} \times \left(\vec{E} + \frac{\partial \vec{A}}{\partial t}\right) = \vec{0}$$

$$\Downarrow$$

$$\vec{E} + \frac{\partial \vec{A}}{\partial t} = -\vec{\nabla} \varphi$$

$$\boxed{\vec{E} = -\vec{\nabla} \varphi - \frac{\partial \vec{A}}{\partial t}}$$

1. Maxw. Gl. $\vec{\nabla} \cdot \vec{E} = -\vec{\nabla}^2 \varphi - \frac{\partial}{\partial t} \vec{\nabla} \cdot \vec{A} = \frac{\rho}{\varepsilon_0}$ (1)

4. Maxw. Gl.: $\vec{\nabla} \times \vec{B} = \vec{\nabla} \times (\vec{\nabla} \times \vec{A}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \vec{\nabla}^2 \vec{A} = \mu_0 \vec{J} + \varepsilon_0 \mu_0 \frac{\partial}{\partial t}\left(-\vec{\nabla} \varphi - \frac{\partial \vec{A}}{\partial t}\right)$ (2)

$$\text{(1)}\quad \vec\nabla^2\varphi + \frac{\partial}{\partial t}\vec\nabla\cdot\vec A = -\frac{\rho}{\varepsilon_0}$$

$$\text{(2)}\quad \vec\nabla^2\vec A - \varepsilon_0\mu_0\frac{\partial^2\vec A}{\partial t^2} - \underbrace{\vec\nabla(\vec\nabla\cdot\vec A) - \varepsilon_0\mu_0\vec\nabla\frac{\partial\varphi}{\partial t}} = -\mu_0\vec J$$

Konstante $c^2 = \frac{1}{\varepsilon_0\mu_0}$ (Lichtgeschwindigkeit^2)

$$\vec\nabla\Big(\underbrace{\vec\nabla\cdot\vec A + \frac{1}{c^2}\frac{\partial\varphi}{\partial t}}_{\to\, 0}\Big)$$

Eichtransformation

$$(*)\quad \begin{bmatrix} \vec A = \vec A' + \vec\nabla\Lambda(\vec r,t) \\ \varphi = \varphi' - \frac{\partial}{\partial t}\Lambda(\vec r,t) \end{bmatrix} \rightarrow$$

$$\vec B = \vec\nabla\times\vec A = \vec\nabla\times\vec A' + \vec 0$$

$$\vec E = -\vec\nabla\varphi - \frac{\partial\vec A}{\partial t} = -\vec\nabla\Big(\varphi' - \cancel{\frac{\partial\Lambda}{\partial t}}\Big) - \frac{\partial\vec A'}{\partial t} - \cancel{\frac{\partial}{\partial t}\vec\nabla\Lambda}$$

neu

$$\vec\nabla\cdot\vec A + \frac{1}{c^2}\frac{\partial\varphi}{\partial t} =$$

alt

$$= \underbrace{\vec\nabla\cdot\vec A' + \frac{1}{c^2}\frac{\partial\varphi'}{\partial t}}_{\text{gegebene Funktion } F(\vec r,t)} + \vec\nabla^2\Lambda - \frac{1}{c^2}\frac{\partial^2\Lambda}{\partial t^2} = 0$$

Lösung der Gleichung

$$\vec\nabla^2\Lambda - \frac{1}{c^2}\frac{\partial^2\Lambda}{\partial t^2} = -F(\vec r,t)$$

Invarianz von $\vec B$ und $\vec E$ $\Rightarrow$ Invarianz der Gln. (1) & (2)

div $\vec E$ rot $\vec B = \ldots \frac{\partial\vec E}{\partial t}$

aber man kann $\Lambda(\vec r,t)$ speziel wählen

Lorenz - Eichung

(Ludwig Lorenz ≠ Hendrik Lorentz)

Lorentz-Kraft
Lorentz - Transformation

$\Lambda(\vec{r},t)$ ist so, dass $\boxed{\vec{\nabla}\cdot\vec{A} + \frac{1}{c^2}\frac{\partial\varphi}{\partial t} = 0}$ $\Rightarrow$ $\frac{\partial}{\partial t}(\vec{\nabla}\cdot\vec{A}) = -\frac{1}{c}\frac{\partial^2\varphi}{\partial t^2}$

1

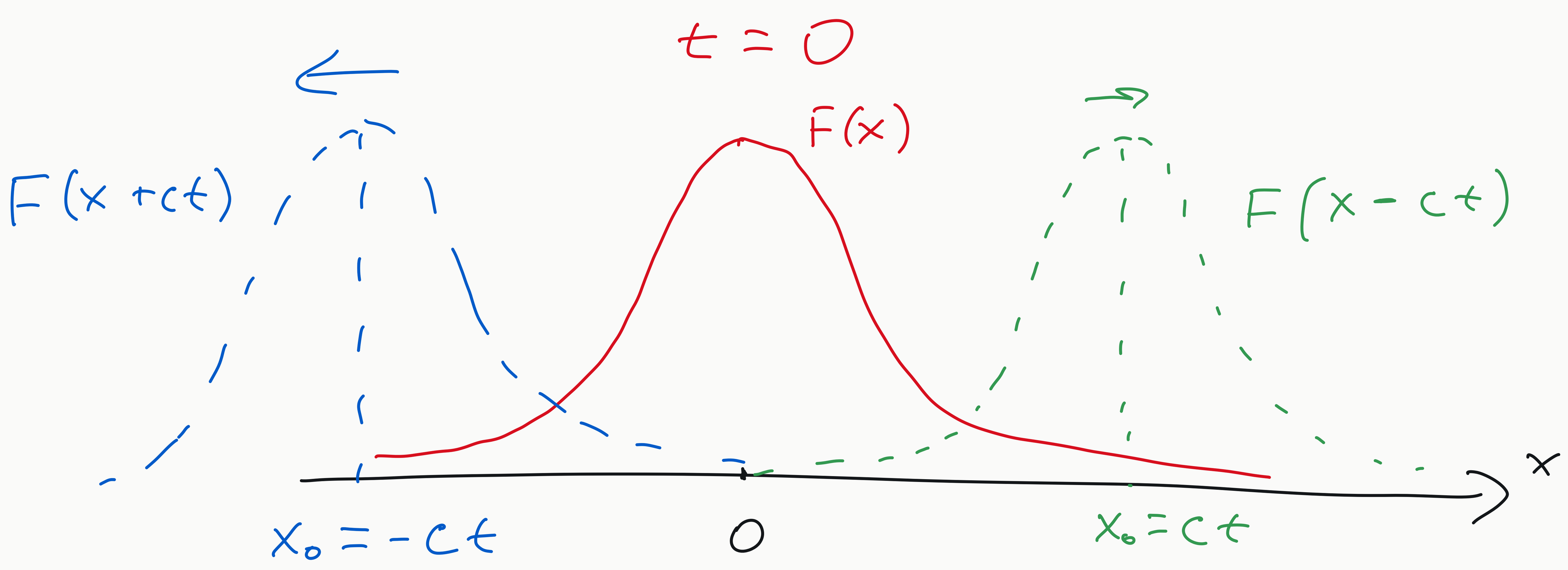
2

$$\boxed{\begin{aligned} \vec{\nabla}^2\varphi - \frac{1}{c^2}\frac{\partial^2\varphi}{\partial t^2} &= -\frac{\rho}{\varepsilon_0} \\ \vec{\nabla}^2\vec{A} - \frac{1}{c^2}\frac{\partial^2\vec{A}}{\partial t^2} &= -\mu_0\vec{J} \end{aligned}}$$

auch die Maxw. Gln. !

Wellengleichung ohne Quellen:

1D: $$\frac{\partial^2 f}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2} = 0$$



Allg. Lösung: $$f(x,t) = F(\underbrace{x \pm ct}_{\xi})$$

Beweis:

$$\frac{\partial f}{\partial x} = \frac{dF}{d\xi}\overset{1}{\overset{\shortparallel}{\frac{\partial \xi}{\partial x}}} = \frac{dF}{d\xi}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{d^2F}{d\xi^2}\frac{\partial \xi}{\partial x} = \frac{d^2F}{d\xi^2}$$

$$\frac{\partial f}{\partial t} = \frac{dF}{d\xi}\underbrace{\frac{\partial \xi}{\partial t}}_{(\pm c)} = \frac{dF}{d\xi}(\pm c)$$

$$\frac{\partial^2 f}{\partial t^2} = (\pm c)\frac{d^2F}{d\xi^2}\underbrace{\frac{\partial \xi}{\partial t}}_{(\pm c)} = (\pm c)(\pm c)\frac{d^2F}{d\xi^2} = c^2\frac{d^2F}{d\xi^2}$$

$$\frac{\partial^2 f}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2} = \frac{d^2F}{d\xi^2} - \frac{1}{\cancel{c^2}}\cancel{c^2}\frac{d^2F}{d\xi^2} = 0$$ ✓

3D: $\vec{\nabla}^2 f - \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2} = 0$

(a) Ebene Welle: $f(x,y,z,t) = F(x \pm ct)$ ← wie in 1D

(b) Kugelsymmetrische Lösung: $f(\vec{r},t) = \frac{1}{r} F(\underbrace{r \pm ct}_{\xi})$

Beweis: $\vec{\nabla}^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial f}{\partial r}\right) + \underbrace{\ldots}_{\text{ahängig von } \frac{\partial}{\partial \theta} \text{ oder } \frac{\partial}{\partial \varphi}}$

$$r^2 \frac{\partial f}{\partial r} = r^2\left(-\frac{1}{r^2}F + \frac{1}{r}\frac{dF}{d\xi}\underbrace{\frac{\partial \xi}{\partial r}}_{=1}\right) = -F + r\frac{dF}{d\xi}$$

$$\frac{1}{r^2}\left(\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right)\right) = \frac{1}{r^2}\left(-\cancel{\frac{dF}{d\xi}\underbrace{\frac{\partial \xi}{\partial r}}_{=1}} + \cancel{\frac{dF}{d\xi}} + r\frac{d^2F}{d\xi^2}\underbrace{\frac{\partial \xi}{\partial r}}_{=1}\right) = \frac{1}{r}\frac{d^2F}{d\xi^2}$$

$$\frac{\partial^2 f}{\partial t^2} = \frac{1}{r}\frac{\partial^2 F}{\partial t^2} \overset{(1D)}{=} \frac{1}{r}c^2\frac{d^2F}{d\xi^2}$$

$$\vec{\nabla}^2 f - \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2} = \frac{1}{r}\frac{d^2F}{d\xi^2} - \cancel{\frac{1}{c^2}}\frac{1}{r}\cancel{c^2}\frac{d^2F}{d\xi^2} = 0 \quad ✓$$

Lösung: Harmonische Welle:

$$f(\vec{r},t) = \mathrm{Re}\left[f_0\, e^{i(\vec{k}\cdot\vec{r} \pm \omega t)} \right] \xrightarrow[\hat{x} \parallel \vec{k}]{} \mathrm{Re}\left[e^{ik(x \pm \frac{\omega}{k} t)} \right]$$ ← Eigenfunktion der Translationssymmetrie

$$\hat{T}_{\Delta\vec{r}}\, \hat{T}_{\Delta t}\, f(\vec{r},t) = e^{i\vec{k}\cdot\Delta\vec{r}}\, e^{\pm i\omega\Delta t}\, \underbrace{f(\vec{r},t)}_{\text{Irreduzibelle Darstellung der Symmetriegruppe}}$$

(ohne Re[ ])

$$\left.\begin{aligned} \frac{\partial f}{\partial \vec{r}} &= i\vec{k} f \\ \vec{\nabla}^2 &= -\vec{k}^2 f \\ \frac{\partial^2 f}{\partial t^2} &= -\omega^2 f \end{aligned}\right] \overset{\text{Welleng!}}{\Longrightarrow} \quad -\vec{k}^2 f + \frac{1}{c^2}\omega^2 f = 0$$

$$\omega^2 = c^2 \vec{k}^2$$

$$\omega = \pm c\, |\vec{k}| \quad \text{(dispersionslos)}$$

Fouriertransformation = Entwicklung nach Eigenfunktionen des Operators $\left(\vec{\nabla}^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \right)$

Zum Weiterlesen:

Jackson 6.2, 6.3